

The project parameters are as follows:

1. Team size: 2 students.
2. Hardware platform:
 - a. ESP-32 WROOM-32
 - b. Arduino Mega 2560 with W5100 Ethernet Shield
 - c. RGB LED ring
 - d. AVT5250 LAN relay board
 - e. Network access point
3. The ESP32 module shall host a web server that provides a web-based user interface for controlling the AVT5250 LAN relay board and the RGB LED ring.
4. The web interface shall support control of at least four relays on the LAN relay board.
5. The web interface shall allow switching all RGB LEDs on and off and configuring their color.